

Research Paper

A boundary-spheropolygon element method for modelling sub-particle stress and particle breakage

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ABSTRACT

We present a boundary-spheropolygon element method that combines the boundary element method (BEM) and the spheropolygon-based discrete element method (SDEM). The interaction between particles is simulated via the SDEM, and the sub-particle stress is calculated by BEM. The framework of BSEM is presented. Then, the accuracy and efficiency of the method are analysed by comparison with both analytical solutions and a well-established finite element method. The results demonstrate that BSEM can efficiently provide instant sub-particle stress for particles with an optimized compromise between computational time and accuracy, thus overcoming most of the disadvantages of existing numerical methods.

1. Introduction

Particle breakage is ubiquitous for granular materials. Breakage is caused by many reasons, such as strong compressive loading, large shear displacement, extreme temperature, dehydration or chemical effects. Laboratory studies have proven that breakage causes changes in particle size distribution [1–4] and particle shape [5,6], which in turn change the state variables, such as density and compressibility, that govern the mechanical behaviour of the bulk granular material. These studies have provided a framework for studying particle breakage. However, due to the limitations in experimental observations, laboratory studies are not able to offer more detailed information about the breakage process.

The discrete element method (DEM), alternatively, provides an insightful way to numerically study the breaking process. It allows the particle breakage to be examined at the grain scale, and the breaking process to be captured at the smallest time scale. The strong and weak force chains can be visualized with DEM, and the breakage of each particle can be traced and analysed. For these reasons, DEM has been widely applied to study particle breakage. Existing methods have been summarized into two paradigms: agglomerates and replaceable particles [7]. Agglomerates [8–12] involve representing individual particles by groups of smaller sub-particles that are bonded together through pre-existing forces. The bonds will be broken or, from the computational perspective, erased once their force reaches a certain threshold. The replaceable particles, alternately, simulate each grain as a single entity and replace them with smaller fragments when the characteristic

stress within the original entity exceeds the particle strength.

The agglomerates can dynamically simulate fracturing processes of individual particles, but the numerical implementation is highly inefficient. This is because simulations require a large number of sub-particles to approach the geometrical shape of particles and obtain reliable results. The fact that the smallest particle size is limited to the size of non-breakage elements compromises its ability to generate a correct particle size distribution (PSD). Unrealistic voids are also inevitably introduced if circular particles are used.

The replaceable particle method is more desirable over agglomerates [7]. The earliest study was conducted by Åström and Herrmann [13] using contact force criteria. Tsoungui et al. [14] proposed a model for crushing circular grains with average stress tensor criteria. They developed a decomposition method to define the failure of an individual grain subjected to an arbitrary set of contact forces. Cantor et al. [15] further modified the method for irregular particles' fragmentation. Eliáš [16] and Gladky & Kuna [17] developed similar stress-based crushable models with three-dimensional polyhedral particles; fragmentation of grains is included by splitting the particles into certain patterns of smaller polyhedrons.

High computational efficiency is the greatest advantage of replaceable particles. Meanwhile, such an approach also avoids the inherent problems in determining the current void ratio for an aggregate of disks, which are porous and have internal voids [7]. However, correct local stress distribution and the conservation of mass cannot be guaranteed if circular particles are adopted. The replacement strategy lacks solid physical meaning. Meanwhile, the reliability and efficiency

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of the simulation are largely compromised if angular particles are adopted, for which the relation of contact forces becomes much more complicated and replacement strategies are poorly studied. It can be observed from the results of these studies [16,17] that the ultimate PSD is quite unsatisfying. This is because the average stress method, which is widely used for circular cases, becomes less valid when shape effects are not negligible. Both breakage initiation and shapes of new fragments are extremely difficult to determine with an averaged stress tensor.

A variety of continuous-discontinuous methods have been proposed to fulfil the breakage requirements [18,19]. The basic idea of these hybrid methods is to combine different types of continuous techniques with DEM. The grain-to-grain interaction is simulated via DEM, and the stress state of an individual particle is calculated by continuous methods using contact force information borrowed from DEM. However, existing methods are still not able to provide results with both high accuracy and reasonable efficiency. For example, the finite-discrete element method [19] can accurately calculate the stress state and breakage surface, but it is computationally very expensive. High-quality results require a high number of elements in the mesh on an individual particle. FEM also bears the same disadvantage of agglomerates that are inherited due to the unbreakable finite elements. Luo et al. [18] proposed a combined scaled boundary finite-discrete element method, which is quite efficient; however, it is unable to provide an accurate stress field in each individual particle, and the calculation for the breakage interface is also oversimplified.

In this manuscript, we present a new method, the boundary-spheropolygon element method (BSEM), which combines the advantages of the continuum and discrete methods. It currently consists of a one-way coupling between SDEM and BEM. The sub-particle stress is calculated through the boundary element method, and the interaction between particles is computed with the spheropolygon-based discrete element method. BSEM only requires the mesh at the boundary of particles and thus simplifies the mesh generation. It uses the fundamental solutions for stress problems. The numerical errors are only generated at the boundary, while the displacement and stress field inside the particles are infinitely differentiable. BSEM has relatively higher efficiency for calculations with a lower number of degrees of freedom, and it is computationally less expensive than FEM for the same accuracy. The irregularity of particles and the contact information are effectively captured through the spheropolygon element method. This new method is intended to be the first step towards the development of a new mechanistic approach for grain breakage based on an evaluation of the sub-particle stress.

The paper is organized as follows. The computational methodology of BSEM is presented in Section 2. The validation and performance of the method are discussed in Section 3 with both analytical solutions and FEM results of Brazilian tests. The single-particle tests for BSEM are conducted through a series of parametric studies on the effect of aspect ratio, coordination number, and heterogeneity of the sub-particle stress. In Section 4, the BSEM is applied to uniaxial compression tests in granular models to investigate the relation between the particle circularity and the breakage resistance. A general discussion and conclusions are presented in Section 5.

2. Boundary-spheropolygon element method (BSEM)

2.1. Boundary element method (BEM)

The boundary element method (BEM) is used as the continuous method to calculate the sub-particle stress. This method subdivides the boundary of the particle under consideration into a series of elements [20]. The solution is sought as a linear combination of fundamental solutions, and their coefficients are solved by satisfying the boundary conditions through boundary integral equations. It has been proved that BEM can effectively solve problems for elasticity. However, BEM still has not been combined with DEM in the way we present here.

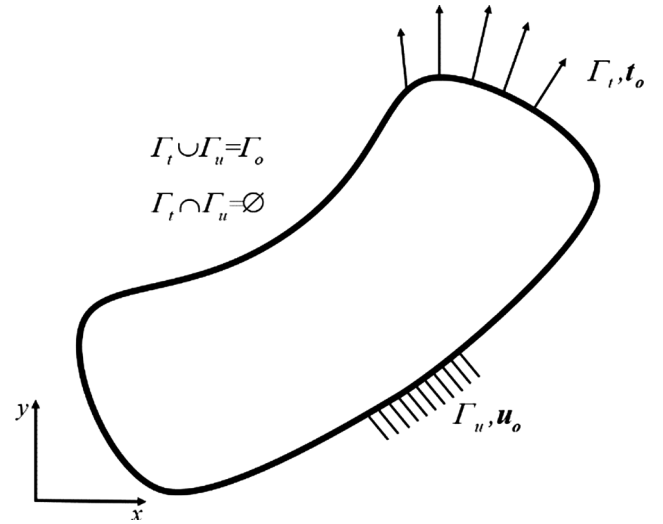


Fig. 1. Definition of boundary conditions in BEM. u_0 and t_0 are the boundary conditions for displacement and traction, respectively. Γ_u and Γ_t denote the surface area for the displacement/traction condition that is known, and they cover the entire boundary of the domain Ω .

In two-dimensional granular flow, the quasi-static sub-particle stress can be regarded as a plane elastostatic problem. Let us consider a homogenous area as shown in Fig. 1. The boundary conditions can be written as:

$$u(x) = u_0, \quad x \in \Gamma_u \quad (\text{Dirichlet boundary conditions}), \quad (1)$$

$$t(x) = t_0, \quad x \in \Gamma_t \quad (\text{Neumann boundary conditions}), \quad (2)$$

where $u = (u_1, u_2)$ and $t = (t_1, t_2)$ denote the displacements and tractions, respectively. The Somigliana's identity is used to express the displacement of the domain point in terms of the boundary conditions. The boundary integral equation in the absence of body forces is expressed as [20,21–23]:

$$c_{ij}u_j(p) = \int_{\Gamma} u_{ij}^*(p, q)t_j(q)d\Gamma - \int_{\Gamma} t_{ij}^*(p, q)u_j(q)d\Gamma \quad (3)$$

where $u_{ij}^*(p, q)$ and $t_{ij}^*(p, q)$ are the Kelvin fundamental solutions. They denote the i th component of displacement and traction at point q induced by the unit load at point p in the j th direction. The coefficients c_{ij} represent the boundary coefficients that can be written as $c_{ij} = \delta_{ij}/2$ with the Kronecker delta δ_{ij} . All of the subscripts ($i, j, k, l = 1, 2$) follow the Einstein notation rule. The Kelvin fundamental solutions for the displacements and boundary tractions for an isotropic body are written as [24,25]:

$$u_{ij}^*(p, q) = \frac{1}{8\pi G(1-\nu)}[(3-4\nu)\ln(\frac{1}{r})\delta_{ij} + r_{,i}r_{,j}] \quad (4)$$

$$t_{ij}^*(p, q) = \frac{1}{4\pi(1-\nu)r}\{r_{,n}[(1-2\nu)\delta_{ij} + 2r_{,i}r_{,j}] - (1-2\nu)(r_{,i}n_j - r_{,j}n_i)\} \quad (5)$$

where G is the shear modulus and ν denotes Poisson's ratio. The scalar r is the Euclidean distance from p to q , and $r_{,i} = (p_i - q_i)/r$ denote the derivatives with respect to p_i . The vector component n_i is the direction cosine of the unit outward normal vector at q . The quantity $r_{,n} = r_{,i}n_i$ represents the derivative of r in the direction of the outward normal at the point p .

2.1.1. Boundary discretization and integration

Based on the boundary integral equation, the boundary can be discretized with a finite number of elements, as shown in Fig. 2. The discretised equation for a boundary element can be written as:

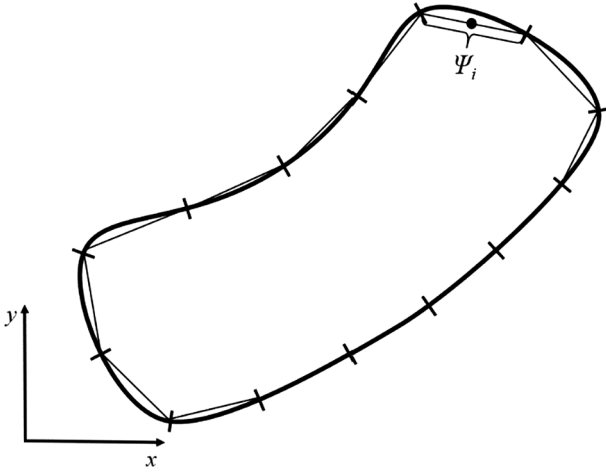


Fig. 2. The boundary of the domain is discretised into a series of piecewise lines, and each line is regarded as a boundary element Ψ_i .

$$c_{ij}u_j(\mathbf{p}) = \sum_{l=1}^N \left[\int_{\Gamma_l} u_{ij}^*(\mathbf{p}, \mathbf{q}) t_j(\mathbf{q}) d\Gamma - \int_{\Gamma_l} t_{ij}^*(\mathbf{p}, \mathbf{q}) u_j(\mathbf{q}) d\Gamma \right] \quad (6)$$

Then, all of the boundary elements form a set of linear equations:

$$\mathbf{c}_m \mathbf{u}_m = \sum_{l=1}^N \left\{ \int_{\Gamma_l} (\mathbf{u}_{lm}^* \mathbf{t}_l - \mathbf{t}_{lm}^* \mathbf{u}_l) d\Gamma \right\} \quad (7)$$

where l and m are the references of boundary elements, and N is the total number of elements ($l, m = 1 \dots N$). The constant element is used to achieve optimal efficiency, which means $\mathbf{u} = u_i$ and $\mathbf{t} = t_i$ ($i = 1, 2$) are constant values within each element.

The matrices of the fundamental solutions are:

$$\mathbf{t}^* = \begin{bmatrix} t_{11}^* & t_{12}^* \\ t_{21}^* & t_{22}^* \end{bmatrix}, \quad \mathbf{u}^* = \begin{bmatrix} u_{11}^* & u_{12}^* \\ u_{21}^* & u_{22}^* \end{bmatrix} \quad (8)$$

$$\mathbf{H}_{lm} = (\mathbf{c}_m + \int_{\Gamma_l} \mathbf{t}_{lm}^* d\Gamma) \quad (9)$$

$$\mathbf{G}_{lm} = \int_{\Gamma_l} \mathbf{u}_{lm}^* d\Gamma \quad (10)$$

Therefore, the matrix for the unknown boundary elements is:

$$\sum_{l=1}^N \mathbf{H}_{lm} \mathbf{u}_l = \sum_{l=1}^N \mathbf{G}_{lm} \mathbf{t}_l \quad (11)$$

$$\mathbf{A} \mathbf{U} = \mathbf{B} \mathbf{T} \quad (12)$$

where $\mathbf{U}^T = [\mathbf{u}_1^T, \mathbf{u}_2^T, \mathbf{u}_3^T, \dots, \mathbf{u}_N^T]$ and $\mathbf{T}^T = [\mathbf{t}_1^T, \mathbf{t}_2^T, \mathbf{t}_3^T, \dots, \mathbf{t}_N^T]$ are the nodal displacements and surface tractions; the coefficient matrices $\mathbf{A}_{2N \times 2N}$ and $\mathbf{B}_{2N \times 2N}$ are constructed as block matrices of the 2×2 matrices $\mathbf{H}$ and $\mathbf{G}$ given by Eqs. (9) and (10). After inverting $\mathbf{A}$ and obtaining all boundary displacements, the stress state at any position of the particle can be calculated as:

$$\sigma_{ij} = \int_{\Gamma} D_{kij} t_k d\Gamma - \int_{\Gamma} S_{kij} u_k d\Gamma \quad (13)$$

where the tensor components of D_{kij} and S_{kij} are given as [20,25]:

$$D_{kij} = \frac{1}{r} \{ (1-2\nu) [\delta_{ki} r_{,j} + \delta_{kj} r_{,i} - \delta_{ij} r_{,k}] + 2r_{,i} r_{,j} r_{,k} \} \frac{1}{4\pi(1-\nu)} \quad (14)$$

$$S_{kij} = \frac{2}{r^2} \left\{ \begin{aligned} & \frac{2}{\partial n} [(1-2\nu) \delta_{ij} r_{,k} + \nu (\delta_{ik} r_{,j} + \delta_{jk} r_{,i}) - 4r_{,i} r_{,j} r_{,k}] \\ & + 2\nu (n_i r_{,j} r_{,k} + n_j r_{,i} r_{,k}) + (1-2\nu) (2n_k r_{,i} r_{,j} + n_j \delta_{ik} + n_i \delta_{jk}) \\ & - (1-4\nu) n_k \delta_{ij} \end{aligned} \right\} \frac{1}{4\pi(1-\nu)} \quad (15)$$

A four-point Gaussian-Legendre quadrature is adopted as the numerical integration method in both Eqs. (9) and (13).

2.1.2. Regularization

The method of *regularization* is a highly effective way to eliminate the rigid-body mode in a coefficient matrix (i.e., matrix $\mathbf{A}$ in Eq. (12)) of elasticity. The major issue in calculating sub-particle stress is that, for most of the particles, all boundary tractions are given while boundary displacements are unknown. Thus, the solution is not unique under pure Neumann boundary conditions. The free rigid-body mode (RBM) is inevitably provoked, and it makes the accuracy of the solution highly compromised or even incorrect. The usual measure for eliminating the RBM is to restrain the body, which requires the introduction of extra conditions. However, restraining is not an ideal method for the particles in DEM simulations where the particles are usually free to exhibit rigid-body movement and rotation. The alternative method of *regularization* is used to eliminate the effect of RBM and obtain the stress field. This method does not require the definition of restraining conditions or any extra procedure of the mesh. The stress components at each time step can be calculated instantly with the DEM simulation.

The regularization method for the non-symmetric matrix was proposed by Lutz [28]. It involves first computing the singular stiffness matrix and then suitably modifying it using linear algebra. Considering the non-symmetric square matrix in $\mathbf{A} \mathbf{U} = \mathbf{B} \mathbf{T}$ (Eq. (9)), the $\mathbf{B}$ matrix must be non-singular since the traction $\mathbf{T}$ is uniquely defined by the displacement $\mathbf{U}$; the null space $\mathbf{N}(\mathbf{A})$ of $\mathbf{A}$, however, is spanned by the rigid-body modes since pure traction conditions are not able to define a unique solution. The basis of $\mathbf{N}(\mathbf{A})$ for the two-dimensional case is written as:

$$\gamma_1^T = [1, 0, 1, 0, 1, 0, \dots, 1, 0]$$

$$\gamma_2^T = [0, 1, 0, 1, 0, 1, \dots, 0, 1]$$

$$\rho^T = [-y_1, x_1, -y_2, x_2, \dots, -y_N, x_N] \quad (16)$$

where γ_1 , γ_2 , and ρ denote two translation vectors and a rotation vector, respectively. The solution for

$$\mathbf{A} \mathbf{x} = \mathbf{b}, \quad (17)$$

can be written as

$$\mathbf{x} = \mathbf{z} + \boldsymbol{\eta} \mathbf{y} \quad (18)$$

where $\mathbf{z}$ is orthogonal to the null space $\xi_i = \gamma_i$ ($i = 1, 2$) and $\xi_i = \rho_i$ ($i = 3$), which consist of the matrix $\boldsymbol{\eta}_{2N \times 3}$; $\mathbf{y}$ is the coefficient matrix. Now, define a new matrix $\mathbf{C}$ as

$$\mathbf{C} = \mathbf{A} + \mathbf{J} \boldsymbol{\eta}^T \quad (19)$$

with $\mathbf{J}_{2N \times 3}$ and the rank of $\mathbf{J}$ equal to three. Obviously,

$$\mathbf{C} \mathbf{z} = (\mathbf{A} + \mathbf{J} \boldsymbol{\eta}^T) \mathbf{z} = \mathbf{A} \mathbf{z} + \mathbf{0} = \mathbf{b} \quad (20)$$

where the vector $\mathbf{z}$ is the solution without the rigid body mode. The form of $\mathbf{J}$ for a nonsymmetric matrix is written as:

$$\mathbf{J} = [\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3], \quad \mathbf{d}_i = \mathbf{B} \xi_i \quad (21)$$

The rigid body mode could also be removed from the solution after solving the matrix $\mathbf{A}$ with a non-zero nullity.

2.2. Spheropolygon discrete element method (SDEM)

The spheropolygon discrete element method (SDEM) is selected as the DEM part of the combined algorithm due to its unique advantages [26,27]. The geometrical irregularity of a particle is properly represented as a Minkowski sum of a polygon with a disk, and the multiple contacts between irregular particles are calculated based on distances between vertices and edges. These features make BSEM computationally efficient. The contact's information can be easily borrowed by the traction vector as boundary conditions in the continuous

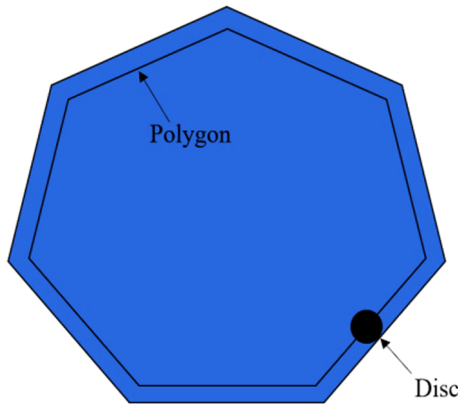


Fig. 3. Spheropolygon element obtained by sweeping a disk around a polygon.

part of BSEM. Most importantly, point contacts are more realistic loading conditions for particles prior to breakage in granular materials.

A spheropolygon is the Minkowski sum of a polygon with a certain number of vertices representing the irregular shape and a disk with radius a , which defines the elastic area for the calculation of contact forces. Formally, the Minkowski sum of two sets of points P and Q of a vector space is given by:

$$P + Q = \{ \mathbf{x} + \mathbf{y} \mid \mathbf{x} \in P, \mathbf{y} \in Q \} \quad (22)$$

This operation is geometrically equivalent to the sweeping of a disk around the profile of the polygon without changing its relative orientation [26]. For example, the irregular shape of the polygon in Fig. 3 can be accurately approximated by a spheropolygon element with a few boundary lines and a disk sweeping its profile. The polygon defines the element shape, and the disk is used for contact calculation. This is a more effective approach than using a cluster of small particles to approach this shape, especially when the particle shape is more complicated. A variety of complex shapes can be generated using spheropolygons, which can properly represent the irregular shape of particles for the simulation of breakage with optimal efficiency.

The interaction force of the spheropolygon is defined by vertex-edge forces. For example, let us consider two spheropolygons SP_i and SP_j with their polygons P_i and P_j and the radii of the disks a_i and a_j . Each polygon is defined by its own set of vertices $\{V\}$ and edges $\{E\}$. The overlapping length between each vertex-edge pair (V, E) is written as:

$$\delta(V, E) = \langle a_i + a_j - d(V, E) \rangle \quad (23)$$

where $d(V, E)$ is the Euclidean distance between the vertex V and the edge E . Therefore, the force applied on particle i by particle j is expressed as:

$$\mathbf{F}_{ij} = -\mathbf{F}_{ji} = \sum_{V_i E_j} \mathbf{F}(V_i, E_j) + \sum_{V_j E_i} \mathbf{F}(V_j, E_i) \quad (24)$$

and the torque on particle i is:

$$\tau_{ij} = \sum_{V_i E_j} (\mathbf{p}(V_i, E_j) - \mathbf{s}_i) \times \mathbf{F}(V_i, E_j) + \sum_{V_j E_i} (\mathbf{p}(V_j, E_i) - \mathbf{s}_i) \times \mathbf{F}(V_j, E_i) \quad (25)$$

where $\mathbf{s}_i$ is the centre of mass of particle i and $\mathbf{p}$ is the point of contact, which is defined as the middle point of the overlap area between the vertex and the edge:

$$\mathbf{p}(V, E) = \mathbf{X} + \left(a_i - \frac{1}{2} \delta(V, E) \right) \frac{\mathbf{X} - \mathbf{Y}}{\|\mathbf{Y} - \mathbf{X}\|} \quad (26)$$

where $\mathbf{X}$ is the position of the vertex V , and $\mathbf{Y}$ is its closest point on the edge E . The movement of the centre of mass $\mathbf{r}_i$ and the orientation φ_i of the particle are governed by the equations of motion:

$$m_i \ddot{\mathbf{s}} = \sum_j \mathbf{F}_{ij}, \quad I_i \ddot{\varphi} = \sum_j \tau_{ij} \quad (27)$$

where m_i and I_i are the mass and moment of inertia of the particle; The linear elastic model is used throughout this paper. The force $\mathbf{F}$ can be written as:

$$\mathbf{F} = k_n \delta_n \mathbf{N} + k_t \delta_t \mathbf{T} \quad (28)$$

where k_n and k_t are the normal and tangential stiffness, respectively; $\mathbf{N}$ and $\mathbf{T}$ are the normal and tangential unit vector; δ_n denotes the length of the overlap; δ_t is the tangential relative displacement between two particles that accounts for frictional forces. It is limited by the condition $k_t \delta_t \leq \mu k_n \delta_n$, where μ is the friction coefficient [26].

2.3. Coupling between BEM and SDEM

The coupling process between the BEM and SDEM is conducted by importing contact forces into the traction vector (Eq. (12)). Currently, the BSEM is a one-way coupling method that only calculates the sub-particle stress but does not consider the breakage of the particle. The fully coupled BSEM should further replace the broken particles [7] at each time step based on their stress state. In this subsection, we introduce the details of the coupling procedure. General considerations for accuracy and computational efficiency are also discussed, which include methods for avoiding the near-boundary singularity and necessary conditions to guarantee the quasi-static state of a particle before conducting the stress calculation.

2.3.1. Coupling through contact forces

In SDEM, the interaction between spheropolygons is calculated from vertex-edge contact forces. Each contact is defined by a vector $\mathbf{f}$ representing the contact force and a vector $\mathbf{p}$ representing the point of the application of the contact force. For a given contact, the particle whose vertex pierces the edge is called the *active particle* and the particle whose edge is pierced is the *passive particle*. The contact force is taken as $\mathbf{f}$ for the passive particle and $-\mathbf{f}$ for the active one. This information needs to be passed to the BEM for the calculation of the sub-particle stress as boundary conditions. Edges of each spheropolygon particle are dilated with a length of the radius of its disk as the new boundary for BEM stress analysis.

Fig. 4 shows the boundary discretization and procedure to import the contact forces into the boundary traction vector. The passive particle in Fig. 4(a) is in contact with three active particles. The intersection points of dilated straight lines, which are parallel to the edges of the original polygon, are used as new vertices. These vertices $\{V^*\}$ define what is called here the “dilated polygon”. In the same way, the edges defined by the new vertices are “dilated edges”. The edge connecting V_1^* to V_2^* is shown in Fig. 4(b). The contact position and force under the global coordinate system (x, y) need to be first transformed to the particle's local coordinate system (ξ_1, ξ_2) using the position and orientation of the particle. The contact force $\mathbf{f}$ is then assigned to the closest boundary element as follows: Linear boundary elements are generated on all the edges, where the first and last node coincides with the two vertices. The contact-bearing edge is selected based on the minimum point-to-line distance as follows: The contact force $\mathbf{f}$ is assigned to the boundary element Ψ_i if $\mathbf{p}$ falls into the element's projection on the ξ_1 axis with the slope $|k| \leq 1$ (or on the ξ_2 axis for $|k| > 1$). The boundary traction at the element Ψ_i is calculated in terms of $\mathbf{f}$ as $\mathbf{t}_i = (t_x = f_x/l_i, t_y = f_y/l_i)$, where l_i is the length of the element.

Fig. 4(c) indicates that $\mathbf{f}$ is entirely assigned to the corresponding element on the closest edge of the passive particle. However, the contact force $-\mathbf{f}$ applied on the active particle is assigned to the closest element of each edge as shown in Fig. 4(d). It is first decomposed as $-\mathbf{f} = f_i \mathbf{n}_1 + f_j \mathbf{n}_2$, where f_i and f_j are the scalars for the force magnitude. $\mathbf{n}_1$ and $\mathbf{n}_2$ are the inner normal unit vector of the corresponding edge under the local coordinate system. Then the decomposed forces $f_i \mathbf{n}_1$ and

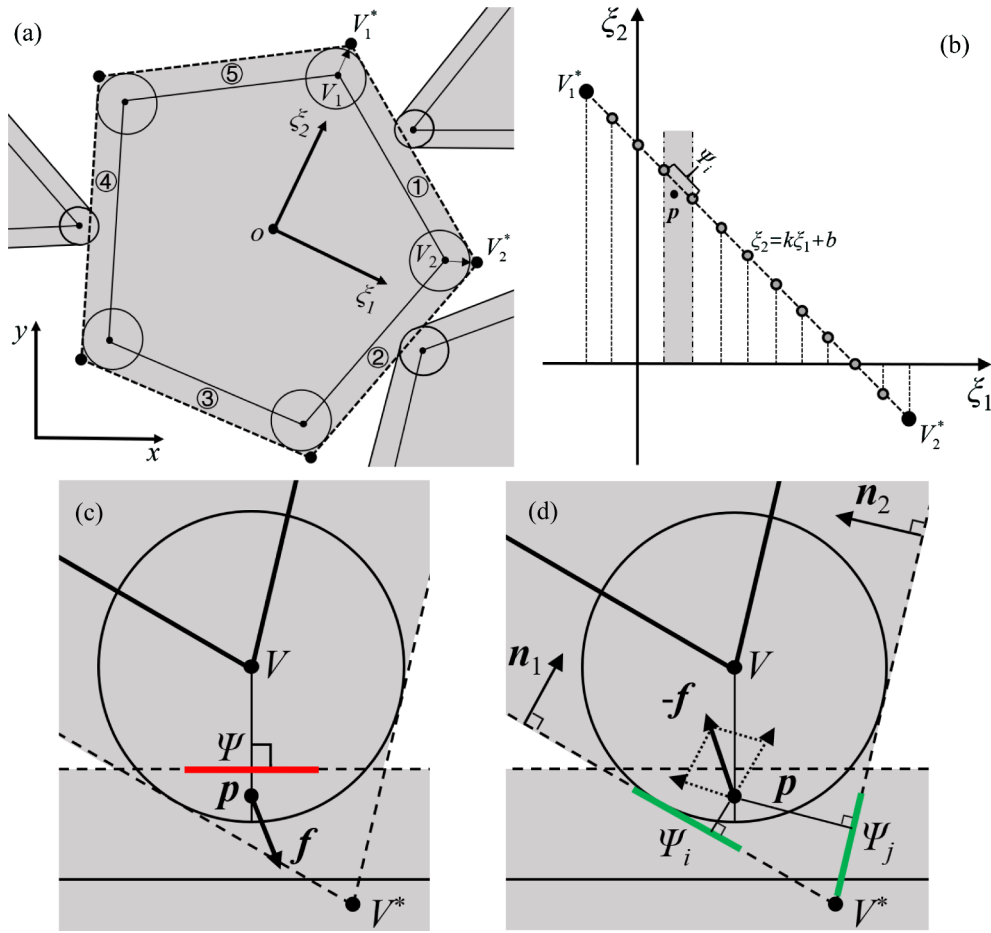


Fig. 4. Procedure for importing SDEM contact forces into the BEM traction vector; (a) particle with three vertex-edge contacts, where the solid and dashed line represent the original and dilated polygon respectively. A “dilated edge” is defined by intersection points (V_1^* , V_2^*) and each dilated edge has an index that is marked with a circle in the figure; (b) The dilated edge connecting V_1^* with V_2^* is parametrized as $\xi_2 = k\xi_1 + b$ using the local coordinates system (ξ_1 , ξ_2), where k is the slope and b is the ξ_2 -axis intercept. The discretization of the edge in boundary element is also shown; (c) the contact force f is entirely assigned to the closest boundary element on the passive particle (in red colour); (d) the contact force $-f$ is decomposed and assigned to the closest element (in green colour) on each edge of the active particle, n_1 and n_2 is the inner unit normal vector for the corresponding edge. The decomposition fulfils the relationship $-f = f_i n_1 + f_j n_2$ where f_i and f_j are the magnitudes of the forces on the elements. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$f_j n_2$ are assigned to the element accordingly using the projection method in Fig. 4(b).

For the disk element, the boundary is discretized with an inscribed polygon (the number of 300 segments is used in this paper). f is assigned to the corresponding element based on the projection of the local polar coordinate system where the origin located at the centre of the disk. The concepts of the passive and active particle are no longer applied.

Each edge has an index β , which represents its sequence on the particle, and an attribute value N_β to represent its number of elements. Each boundary element has two indexes: I_r and I_g . I_r represents the relative index of an element on the edge. This index is used to determine the closest element on the edge based on the projection of the contact position. The index I_g denotes the global index of an element in the traction vector T in Eq. (12), which is the absolute index of the element on the particle. The relationship among the indexes can be written as

$$I_g = I_r + \sum_{k=1}^{\beta-1} N_k \quad (29)$$

If the contact position coincides with a node between two elements, the contact force will be equally split and assigned to both elements. All indexes are generated in a clockwise sequence under the local coordinate system of each particle in this paper.

The discontinuous method (SDEM) and the continuous method

(BEM) are combined through the contact information. The position, velocity and contact forces of particles are simulated by SDEM, while the sub-particle stress is calculated by BEM using contact forces as boundary traction. The information of stress states could be further used to determine the damaged area and the shape of new particles for the replacement.

2.3.2. Avoid the near-boundary singularity

The *near-boundary singularity* refers to the divergences that occur when an integral point of BEM approaches the boundary. The error of numerical integrands becomes quite large over the length of the element [29]. This problem is caused by fundamental solutions, which involve distance-dependent terms such as $\ln(1/r)$, $1/r$, and $1/r^2$. There are two methods to avoid or minimize such a problem. The first approach is to use a parameterized Gaussian quadrature method [29,30] when a region point is approaching the boundary. The other method is to properly ignore the near-boundary area through the particle dilation process of the SDEM. Based on biaxial test experiments, several authors [10,11,31] conclude that the breakage is mainly caused by tensile failures inside the particle. Therefore, the exact values of boundary displacement may not be crucial for particle fragmentation.

As discussed earlier, SDEM requires the polygon, which is the rigid part of the particle that defines its shape, to be dilated by the half-diameter size of its disk so that an elastic contact layer can be created

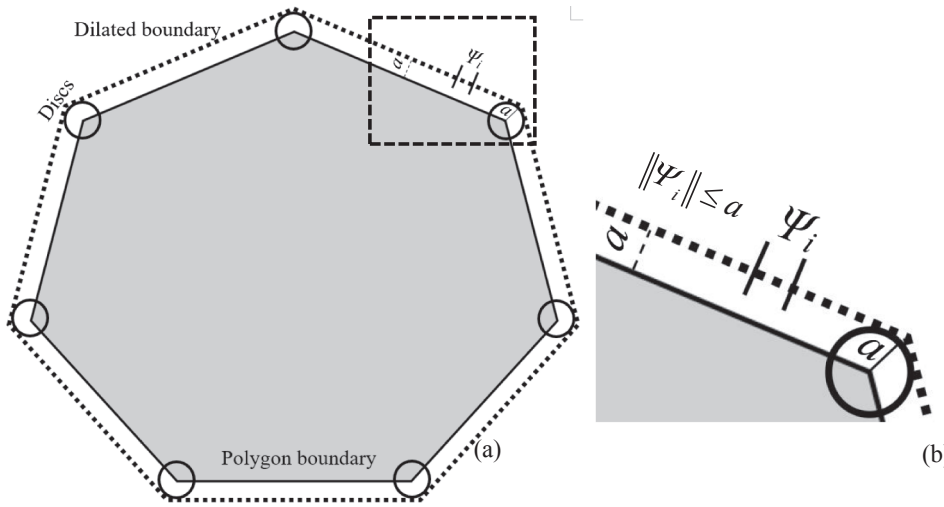


Fig. 5. The particle dilation method is used to avoid the near-boundary singularity. (a) The Minkowski sum of the polygon and the disk of radius a ; (b) enlarged square area in (a) at the boundary. If the length of the element Ψ_i is less than or equal to a , the effect of the near-boundary singularity inside the polygon can be reduced.

with its disc. Alarcon et al. [20] indicated that the near-field singularity of the integrand vanishes at approximately the half-distance of an element. To reduce the effect of the near-field singularity, this strategy is incorporated through particle dilation in BSEM as follows: (i) the length of the element is chosen equal to the radius of the spheropolygon disc, (ii) the boundary for integration is dilated from the rigid one (polygon) to the elastic boundary (polygon + radius), and (iii) only the stress field within the polygon is trusted and used for breakage calculation. Fig. 5 shows how the polygon is dilated for the calculation of sub-particle stress; the boundary discretization is conducted at the dilated boundary. It needs to be noted that dilation is not conducted for the disk element, because the disk element does not need dilation to calculate the contact like the spheropolygon does. Therefore, the trusted area of stress is shrunk with a half-length of the boundary element.

By using particle dilation in BSEM and calculating the stress in the polygon only is computationally more efficient than the parameterized Gaussian quadrature to avoid the near-boundary singularity. Although the actual attrition and erosion near the particle boundary may not be accurately simulated, the fragmentation of particles can be properly captured through the sub-particle stress fields inside the polygon.

2.3.3. Criteria for the evaluation of sub-particle stress

Special conditions need to be fulfilled since SDEM is a dynamic method, while the BEM formulas we used for sub-particle stress calculation are only valid for quasi-static conditions. To guarantee the validity of the BEM calculation of the sub-particle stress, the particle should be in an equilibrium of force and its linear and angular velocity need to be approximately zero. Therefore, the contact forces and velocities need to be examined before the calculation.

The two inequalities, Eqs. (30) and (31), are used to guarantee a static state condition:

$$F_b = \left[\sqrt{\left(\sum_{i=1}^n f_x^i \right)^2 + \left(\sum_{i=1}^n f_y^i \right)^2} \leq e_f \right] \wedge (|\tau_b| \leq e_\tau) \quad (30)$$

where the absolute values of the resultant force F_b and torque τ_b are required to be smaller than the numerical tolerances e_f and e_τ . Ideally, e_f and e_τ are equal to zero. In our simulations $e_f = 1 \times 10^{-6} a_{min} \times k_n$ and $e_\tau = (1 \times 10^{-3} a_{min})^2 \times k_t$ are scaled with contact properties (k_n , k_t), and the minimum radius a_{min} to tolerate the computational errors (e.g., 1×10^{-6} for single-precision variables). This condition guarantees a balanced force and torque on an individual particle. The BEM stress calculation could not provide correct results if this condition is not satisfied.

The second quasi-static condition is imposed on the kinematics:

$$(\sqrt{v_x^2 + v_y^2} \leq e_v) \wedge (v_a \leq e_{va}) \quad (31)$$

This inequality states that both velocity (v_x and v_y) and angular velocity (v_a) should be smaller than $e_v = 1 \times 10^{-4}$ m/s and $e_{va} = 1 \times 10^{-4}$ rad/s. The values are also used as the tolerances of numerical errors. Because regularization is conducted, the results of sub-particle stress should not be affected by the absence of this condition. However, this condition is proposed because the BEM formula used here does not deal with a dynamic situation.

3. Single-particle test results

The BEM and BSEM share the same procedure for the single-particle stress calculation, except where boundary traction is applied directly to the traction vector (Eq. (12)) instead of using inter-particle contact forces. Therefore, the accuracy and the efficiency of BEM, as the method for stress computation of BSEM, are tested in this Section. The classic 2D Brazilian test is first presented, and the result of BEM is compared against the analytical solution. Then, computational efficiency is compared with FEM results. Next, a uniaxial compression test is executed with both circular and elliptic particles, and the full stress fields are obtained to demonstrate the capability of BSEM to calculate the stress state of irregular particles. Both force and torque equilibrium are fulfilled in each test. Only maximum tensile stress is measured and discussed. The results illustrate how abundant and insightful the sub-particle stress information that the well-established BEM can provide is.

3.1. Validation

A Brazilian test is chosen as a benchmark test. It is a widely recognized experimental method to measure the tensile stress of a rock specimen [32]. The classical Brazilian test involves uniaxial loads over a cylindrical sample. The analytical solution of the Brazilian test is well known. Its stress field shows that the maximum tensile stress occurs at the centre of the sample where the crack is initiated as a tensile failure. Based on the experimental data of single grain compression, authors [4,7,10,33] suggest that the characteristic tensile stress induced by a pair of diametrical loads in a spherical grain is consistent with the maximum tensile stress of the Brazilian test. DEM breakage simulations also use the averaged stress tensor to approximate this result of the Brazilian test as a validation for the opposite-loading case [14].

The particle radius $R = 0.1$ m is used in the calculations below. The material properties are as follows: shear modulus $G = 2.0 \times 10^3$ MPa and Poisson's ratio $\nu = 0.2$. The same parameters are applied in all simulations.

The standard Brazilian test is shown in Fig. 6. The compressive

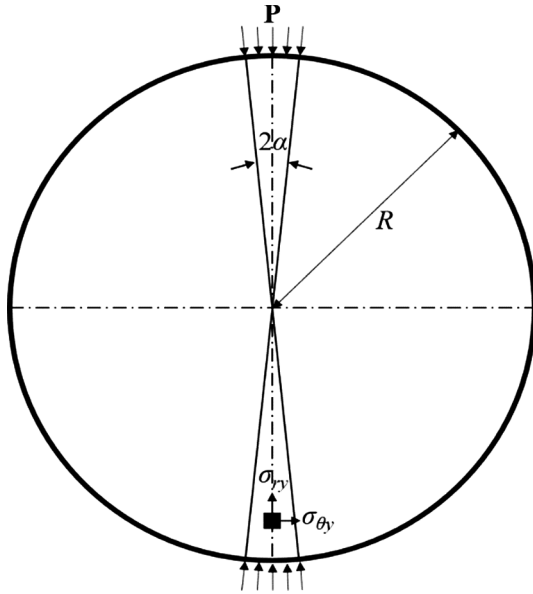


Fig. 6. Brazilian disk test under a uniformly distributed load over finite arcs [32].

pressure P is applied at the top and bottom of the sample. To prevent the sliding of the sample, the loading condition is usually not a pair of perfect point loads but rather a small contact area defined by an angle α .

The results of the BEM calculation are presented in Fig. 7. BEM uses 224 linear boundary elements with a contact angle $\alpha = 7.5^\circ$. The resolution of inner stress points is 200 evenly distributed in the circumscribed square (or rectangle) of the particles, which is used throughout this paper. As we can see, the horizontal and vertical zero lines of shear stress are crossing the centre and are perpendicular to each other. The tensile zone in the middle of the disk and the stress field near the contact areas are also clearly presented. The BEM stress field shows good agreement with the analytical solutions. The tangential normal stress $\sigma_{\theta y}$ and radial normal stress σ_{ry} (in Fig. 6), as a function of the axial coordinate r_y , are given by [32]:

$$\sigma_{\theta y} = \frac{2P}{\pi} \left\{ \frac{(1 - r_y^2/R^2)\sin 2\alpha}{1 - 2r_y^2/R^2 \cos 2\alpha + r_y^4/R^4} - \tan^{-1} \left[\left(\frac{1 + r_y^2/R^2}{1 - r_y^2/R^2} \right) \tan \alpha \right] \right\} \quad (32)$$

$$\sigma_{ry} = -\frac{2P}{\pi} \left\{ \frac{(1 - r_y^2/R^2)\sin 2\alpha}{1 - 2r_y^2/R^2 \cos 2\alpha + r_y^4/R^4} + \tan^{-1} \left[\left(\frac{1 + r_y^2/R^2}{1 - r_y^2/R^2} \right) \tan \alpha \right] \right\} \quad (33)$$

The maximum tensile stress is obtained at the centre of the sample.

Comparison of axial stress components is shown in Fig. 8. The numerical results provided by BEM agree well with the analytical solution

near the central area, while numerical errors can be observed near the boundary area as a result of the near-boundary singularity.

3.2. Performance analysis

The accuracy and efficiency comparisons between BEM and FEM using a quadratic triangular element are shown in Fig. 9. The error is calculated by comparing the tensile stress at the centre of the disk with that of the analytical solution.

The performance of both cases is obtained from the Brazilian test with an increasing number of mesh (boundary and domain) elements. The computational time includes the time consumption of assembling and solving the matrix with *LU decomposition*. It can be observed from Fig. 9(a) that, for the same accuracy, BEM has better efficiency than FEM. and for an equal consumption of computational time, BEM is more accurate. Such difference exists since the computational time is directly linked to the size of the coefficient matrix and the number of degrees of freedom (DoF). It indicates that BEM is a more suitable method for problems with a low number of DoF. To achieve equal accuracy, the smaller size of the matrix (i.e., a lower number of DoF) is more favourable. In this case, particles do not need to be finely meshed, while a continuous and accurate stress field can still be obtained. This advantage of BEM is more significant if the simulation involves a larger number of particles.

It must be pointed out that this advantage of BEM does not stand for high-DoF quasi-static elasticity problem. This is because the stiffness matrix of FEM analysis is sparse, symmetrical, and positive definite, which allows the use of more efficient solvers such as the iterative solver. The iterative solver effectively decreases the computational complexity and compensate for the disadvantage of large matrix size.

3.3. Sub-particle stress calculation

A series of parametric studies of sub-particle stress are presented in this section. The effects of two variables for the breakage, aspect ratio (AR) and coordination number (CN), are investigated with BEM. The sub-particle stress fields are calculated and analysed. The heterogeneity of the sub-particle stress for both circular (the same model used for the Brazilian test) and elongated elliptical grains (the number of linear boundary elements ranged between 230 and 250) are compared. Then, the validity of three formulas for particle strength is examined: the effect of aspect ratio, cushioning effect, and averaged stress tensor. The purpose of this section is to provide a new perspective on the influence of particle shape and meanwhile demonstrate the limitation of the stress averaging method, which is widely accepted and adopted in numerous studies of particle fragmentation.

3.3.1. Aspect ratio

After particle size, the second most important geometrical variable

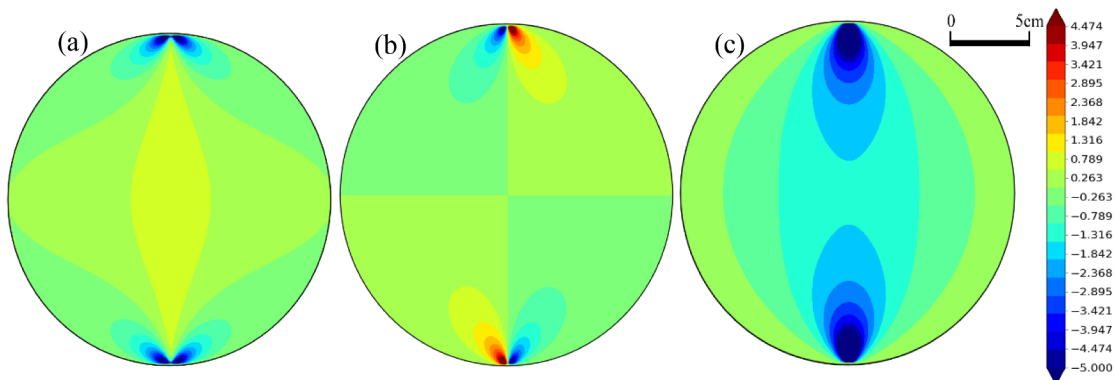


Fig. 7. Stress components of the Brazilian test (Units: MPa); (a) σ_{xx} , (b) τ_{xy} , (c) σ_{yy} .

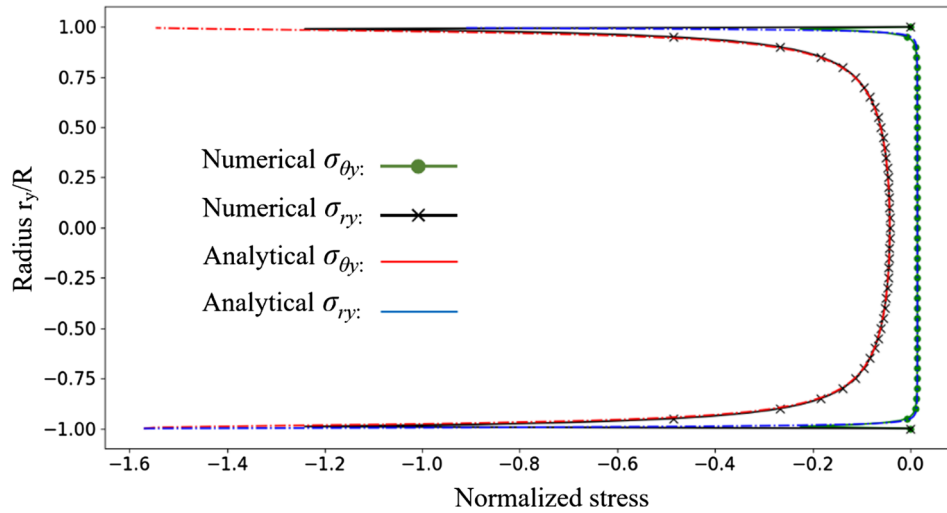


Fig. 8. Axial stress distribution in the Brazilian test. The solid line denotes the analytical solution, and the dashed lines represent results from BEM.

that affects the breakage of a particle is aspect ratio (AR). The characteristic tensile stress σ for a circular grain is a formula of diametric force F and its radius R , which can be written as [4,33]:

$$\sigma = \frac{F}{R^2} \quad (34)$$

As mentioned earlier, this formula is consistent with the analytical solution of maximum tensile stress in a Brazilian test and can be used for the measure of the tensile strength of grains. If the particle is non-circular, Eq. (34) needs to be modified. Based on experimental data, Afshar et al. [31] proposed the following equation:

$$\sigma = \frac{F \times AR}{D^2} \quad (35)$$

where $AR = D/d$ is the aspect ratio; D denotes the sieve-measured diameter of the particle, and d is the laminar thickness. This formula suggests that, with an equal pair of loads, particles with larger AR would have larger tensile stress. Thus, the tensile strength of elongated particles measured or observed would appear to be lower. This modification of the strength formula was proposed by empirical research without providing a theoretical explanation.

With the aim of testing Eq. (35), we conducted BEM tests on a single particle using spheres and ellipses with different aspect ratios. AR is defined here as the ratio of the length of the major and minor axes. $AR = 1.0$ is chosen for circular particles with radius $R = 0.1$ m, and $AR > 1.0$ is used for all the elliptical particles. These geometrical parameters are applied to all of the relevant tests. d is the distance between two loading points; the areas of the particles are set to be equal

to eliminate the statistical effect of the Weibull distribution [33]. Particles are equally loaded with a pair of axial compression pressures, which resembles the Brazilian test. The maximum tensile stress is measured for each case. The diametrical load is equal to 2×10^5 N.

The curve in Fig. 10(a) shows that the maximum stress in a particle increases with AR . The elliptical particles with larger AR are easier to break than circular particles. It is not reasonable that the macroscopic strength can be changed by the shape. As the results suggest, the so-called AR -strength effect is caused by the maximum tensile stress, which is directly related to the shape of the particle. Eq. (35) is thus a linear approximation of the dependency of tensile stress with the aspect ratio of the particle, but non-linearities should be expected.

The second test is conducted to further understand this effect: the same loading condition is applied at different positions of an elliptical particle with $AR = 2.0$, and the loading positions move from the centre towards the tip of the particle with abscissa $x = 0, b/4, b/2, 3b/4$. Fig. 11(b) indicates that the maximum tensile stress increases with the decrease of the distance between the two contact points. It can be well explained with fundamental solutions where the intensity of the stress field is strongly affected by the distance-dependent terms $\ln(1/r)$, $1/r$ or $1/r^2$. For a circular particle, two stress fields overlap with each other and produce the maximum tensile stress at the exact centre. Such overlapping becomes stronger if the distance decreases on particles that have larger AR values or one particle but at different positions. Although the physical meaning is not accurate, the modification in Eq. (35) is consistent with fundamental solutions. It is, therefore, reasonable to use laminar thickness to modify the formula for the strength of a

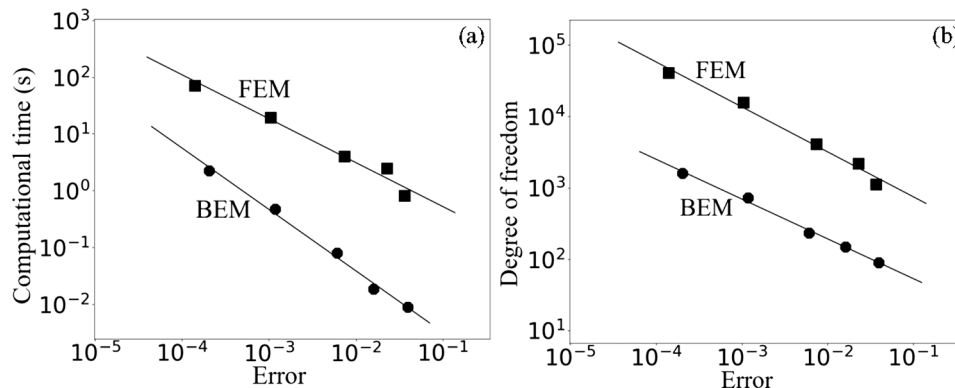


Fig. 9. Computational performance of BEM compared to FEM; (a) computational time versus numerical error; (b) degree of freedom versus numerical error. The solid lines are generated with first-order polynomial regression. BEM uses constant elements, whereas FEM uses iso-parametric quadratic triangular elements.

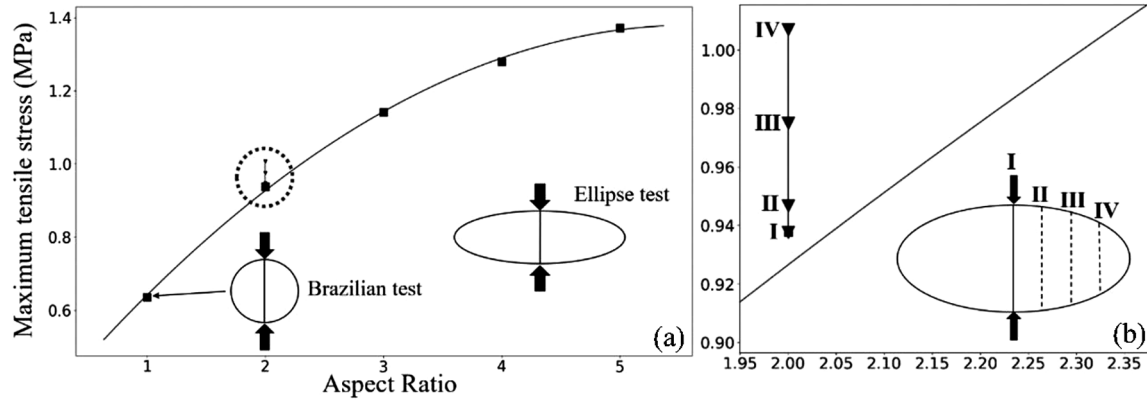


Fig. 10. (a) The maximum variation of the stress with aspect ratio using second-order polynomial regression; $AR = 1.0$ is a 2D Brazilian test; $AR > 1.0$ for tests with elliptical particles; (b) amplified circular area of $AR = 2.0$ with different loading positions.

particle with $AR > 1.0$.

3.3.2. Coordination number

Particle breakage is related to the loading condition, which includes the following three key variables: the load intensity, the central angle of applied loads, and the contact number. The latter is also called the coordination number (CN). McDowell et al. [4] assumed that increasing CN further decreases the chance of breakage. Although they acknowledged that such an assumption is insufficient or even invalid for elongated particles, it has been embedded into many DEM or analytical models of particle breakage. It is reasonable to assume that a large value of CN (e.g., $CN > 10$) may prevent the breakage of a particle due to the cushioning effect [4] (many small particles are in contact with one big particle such as a cushion). However, for a low number of contacts, such an assumption is questionable and requires further examination.

A series of BEM tests are conducted to investigate the effect of low CN on the sub-particle stress. As shown in Fig. 12, circular ($AR = 1.0$) and elliptical particles ($AR = 2.0$) are used for the comparison between different AR s. Four groups of contact angles are applied with $CN = 3, 4$.

The stress fields in Fig. 13 show that the sub-particle stress has a strong dependency on the loading angle. For the circular particle, the maximum overlapping area shrinks with increasing loading angles, which also causes changes of tensile failure position and maximum value.

It can be observed from Fig. 14 that the effect of the coordination number agrees well with the existing assumption regarding the cushioning effect for circular particles. $CN = 4$ has a lower maximum tensile stress than that of $CN = 3$; the drop of maximum tensile stress with the

contact angles is also consistent with it. However, the above is not valid for the elliptical particles, where the shape of the particle distorts stress fields. The maximum tensile stress even increases with contact angle or the value of CN . Such inconsistency is directly caused by the shape of the particle and usually fails to be properly considered by many studies.

3.3.3. Heterogeneity of sub-particle stress

The average stress tensor is the foundation for many of the numerical and analytical methods. It can represent the stress state in a particle without consideration of its irregular geometry. As we discussed in the earlier parts through the analysis of maximum tensile stress, this tensor shows good agreement with the Brazilian test in circular particles, but it is unable to accurately capture the stress state in the elliptical particle. The heterogeneity of the sub-particle stress is further investigated here using circular and elliptical models ($AR = 2.0$) subject to an equal pair of loads. The maximum tensile stress along the different diametrical directions is measured.

The maximum principal stress is plotted diametrically (r_ϕ is the distance to the centre, and R is the distance between the centre and the boundary) at the circular and elliptical particles with different angles. Fig. 15 shows that, for the elliptical particle, the heterogeneity of the stress is stronger than that of the circular particle. The maximum tensile stress can still be found at the central line but is no longer located at its centre point. It also shows a more drastic variation along some of the diametrical directions with a range that includes both negative and positive values. Therefore, the validity of an average tensor is highly compromised for the irregular particle simulation. Neither the breakage time nor the breakage shape can be accurately predicted by using the averaging method [15].

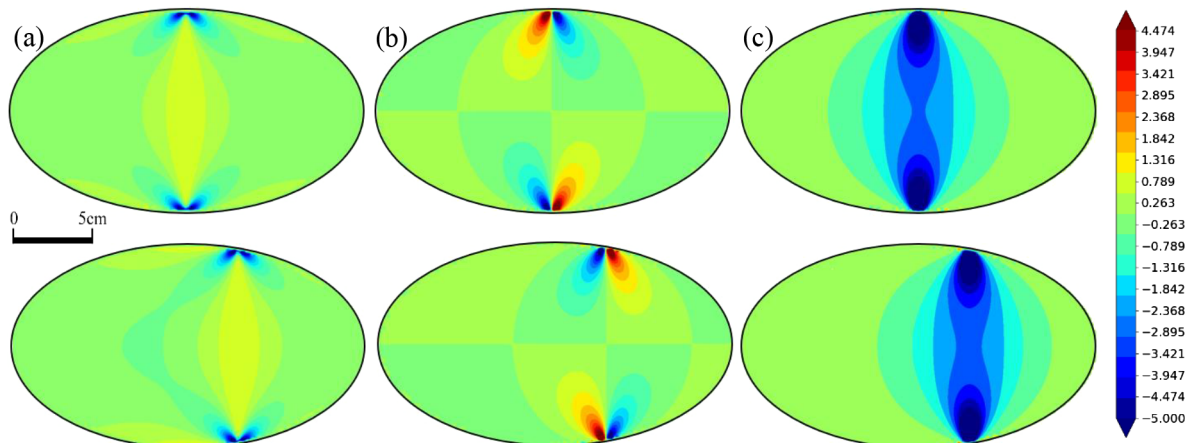


Fig. 11. Stress fields of an elliptical particle with $AR = 2.0$ and the loading positions I and II (Units: MPa); (a) σ_{xx} , (b) τ_{xy} , (c) σ_{yy} .

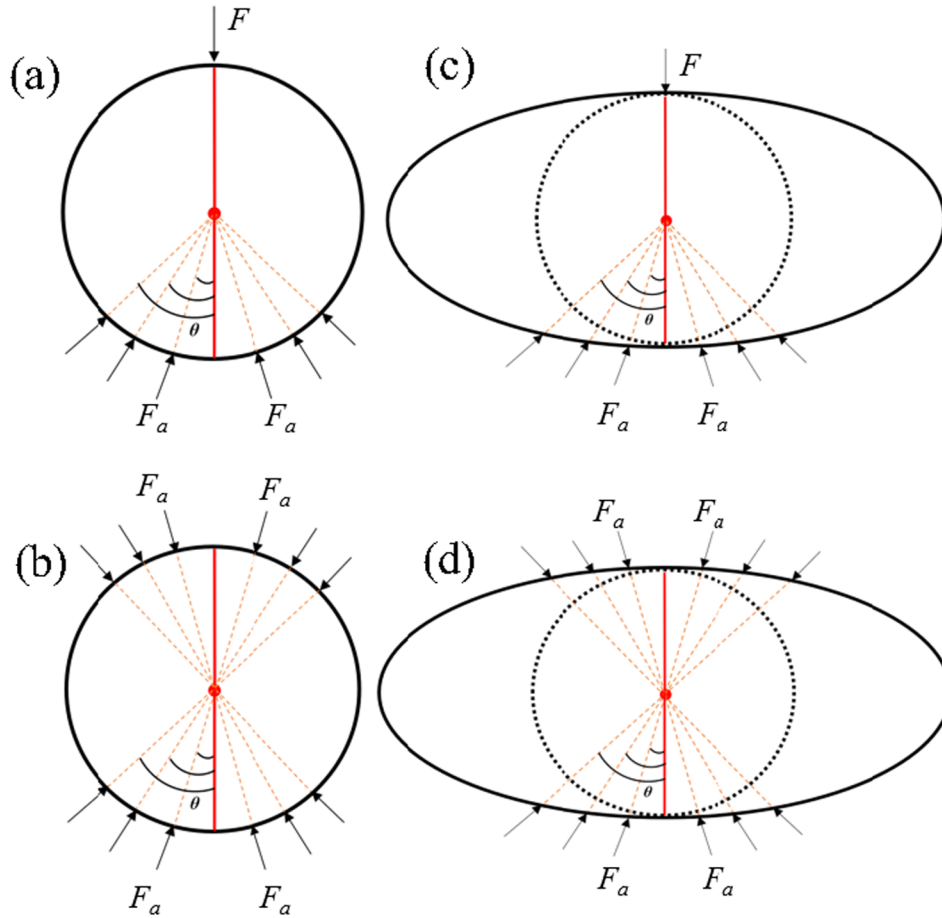


Fig. 12. The loading conditions of circular and elliptical particles, where the imposed loads are: $F = 2 \times 10^5$ N, $F_x = \pm 0.5F \tan(\theta)$, $F_y = -0.5F$, $F_a = (F_x^2 + F_y^2)^{1/2}$, $\theta = 0^\circ, 15^\circ, 30^\circ, 45^\circ$. (a) and (b) $CN = 3$; (c) and (d): $CN = 4$.

3.3.4. Discussion

The analysis of the tensile stress in the previous section provides three important highlights regarding how the description of the particle breakage can be improved, as follows: (1) the aspect ratio is an essential particle shape parameter to include in the particle strength criteria; (2) the increase of the coordination number does not necessarily lead to a reduction of particle breakage as the cushioning effect suggests; and (3) the average stress tensor is not sufficient to predict breakage, and the stress heterogeneities need to be taken into account. The breakage of an individual particle is a complicated process that is affected by a variety of conditions, such as contact forces, material properties, and geometry. The sub-particle stress components govern the distribution of damaged areas and the development of breakage interfaces, and they cannot be reduced to a universal equation. It is not accurate enough to simply average the overall stress states of a particle and ignore the heterogeneity of the stress field. Therefore, the sub-particle stress provides a crucial perspective for a better understanding of particle breakage. It is desirable for a numerical simulation of particle breakage to be governed by accurate stress information.

4. Granular model tests of BSEM

In this section, we use a one-way BSEM coupling method to simulate the uniaxial compression test of granular materials. The tests are performed to demonstrate the ability of BSEM to provide insightful information on sub-particle stress distribution in a granular system. For all of the tests, particles are placed in a container that consists of three rigid walls at right, left and bottom. The compression force is applied by a rigid bar with a constant magnitude $F_y = -2 \times 10^5$ N. The

calculation of sub-particle stress is conducted after the force equilibrium (Eq. (30)) on each particle is achieved, and the entire granular system is in a quasi-static state (Eq. (31)). The specific number of boundary elements on each particle is proportional to its perimeter. This is due to the rule that the length of the element is equal to the radius of the spheropolygon, as discussed in Section 2.3.2. Using a sphero-radius of $a = 1.0 \times 10^{-3}$ m, this rule generates less than 300 elements on the particles simulated here. The gravity is excluded from both the calculation of sub-particle stress and the SDEM simulation. The material properties and parameters of the simulations are listed in Table 1.

Fig. 16 shows how BSEM is conducted for circular and angular particles. It can be seen from the continuity of the stress fields that the contact information provided by SDEM is properly coupled with the traction boundary conditions in BEM. In particular, the stress fields near the contact point of two particles are symmetrical. This indicates that both the magnitude and the position of the contacts are correctly imported into the traction vector and therefore provide a reliable result for the stress field on each particle. Furthermore, the stress fields for angular particles show a stronger concentration and heterogeneity of the contact forces than those of circular particles.

With the aim of further investigating the effect of particle circularity, we perform different simulations by keeping the void ratio approximately constant and only changing the particle circularity. Simulations are performed with disks and regular polygons (dodecagons, decagons, and octagons). The following formula for the circularity is adopted:

$$CR = \frac{4\pi A}{B^2} \quad (36)$$

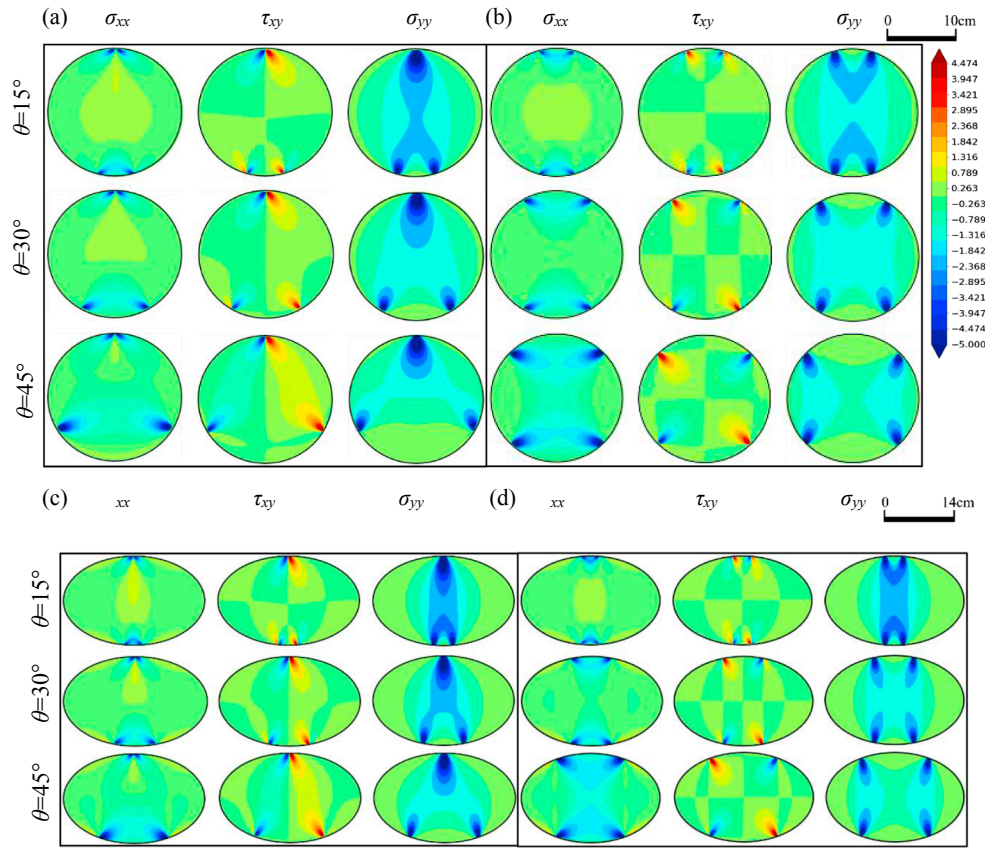


Fig. 13. Stress components for each loading case (a)–(d) in Fig. 12 (Units: MPa). The results of different loading angles are arranged as $\theta = 15^\circ, 30^\circ, 45^\circ$ from top to bottom and stress $\sigma_{xx}, \tau_{xy}, \sigma_{yy}$ from left to right.

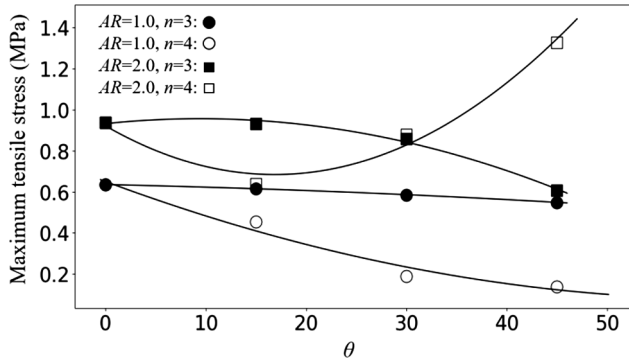


Fig. 14. The maximum tensile stress with different loading angles and coordination numbers.

where A is the area, which is controlled as a constant value ($A = 4\pi$ for the circular particle's radius $R = 2.0$ cm) for all of the particles; B is the perimeter of the particle, which will increase with the decrease of circularity. CR equals 1.0 for a circular particle and decreases with the increase of the number of angles. Only three layers of particles are used to fully exhibit the details of the sub-particle stress fields. A spheroradius of $a = 1.0 \times 10^{-3}$ m is also applied here and produces less than 200 elements for each particle. Particles are arranged with random orientations at the beginning of each simulation to further minimize the influence of the packing pattern.

It can be observed from the sub-particle stress fields in Fig. 17 that the loading forces for circular particles are symmetrical. This is because the compressive load from the bar is evenly distributed to the particles. Particularly, the stress component σ_{yy} has the highest magnitude for each particle as a reaction to the vertical compression. The stress distribution is consistent with the results provided by Zheng and Tannant

[12], who used force chains of agglomerates to represent the sub-particle stress. The results are also consistent with the loading test on single circular particles in Section 4.2, and it further confirms that the sub-particle stress fields in the granular material cannot be represented by the average stress tensor, especially when particles have multiple contacts.

The patterns of sub-particles stress fields for angular particles in Fig. 17 show that the decrease of the circularity enhances the heterogeneity of the contact network. Additionally, such a decrease causes the loading conditions on particles to become less even and no longer symmetrical. Some of the particles bear most of the loads, while the others are much less or not loaded. The stress component σ_{yy} still has the highest value as a reaction to the vertical compression, yet the magnitudes of σ_{xx} and τ_{xy} are increased due to the uneven distribution of forces and changes of particle geometry. The numerical comparison between the results of different circularity (CR) models also confirms that the heterogeneity of the force distribution increases with the decrease of circularity.

The values of the maximum principal stress field for the particles in each model are shown in Fig. 18; the peak value (PV) and averaged variation of the maximum principal stress (AVM) of the entire granular packing is shown in Fig. 19. The stress field in the system of the circular particle is symmetrical, and it also has the lowest variation and peak value. Both PV and AVM are largely increased with the decrease of CR . Generally, the circularity of the particles directly affects the distribution of the force chain. The heterogeneity of the force chain makes the particles unevenly loaded and produces complicated sub-particle stress fields. In such a situation, more breakage will be observed in the model with lower circularity. It must be pointed out that the circularity of the particle does not directly decide the breakage resistance for an individual particle but instead changes the heterogeneity of the force chain in a granular assembly and produces more breakage.

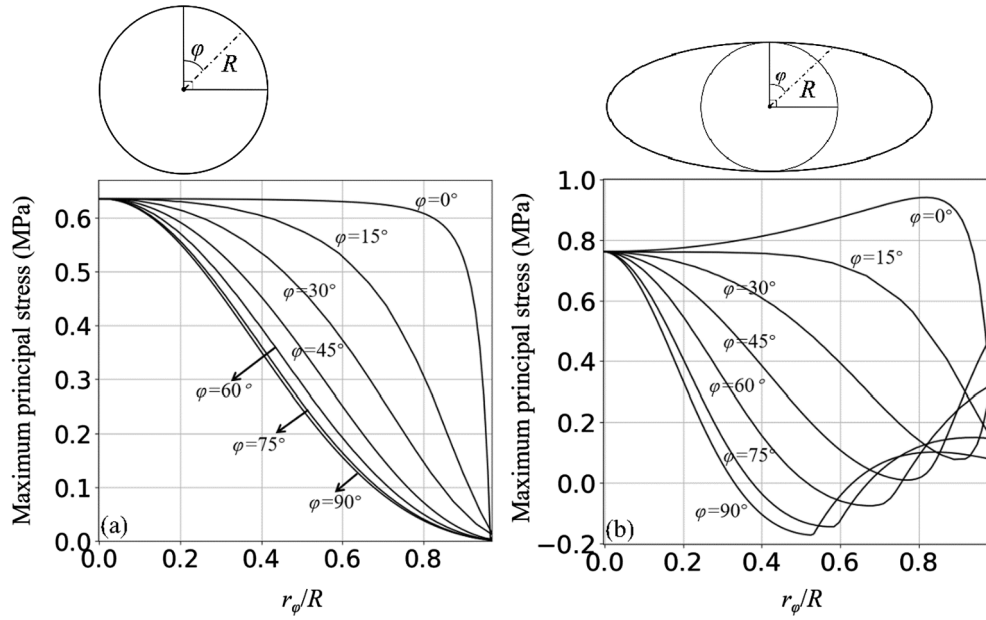


Fig. 15. Maximum principal stress distribution at diametrically opposed directions with different angles; (a) circular particle case, (b) elliptical particle ($AR = 2.0$) case.

Table 1

Parameters used in the simulation.

k_n	Normal stiffness	6.0×10^8	N/m
k_t	Tangential stiffness	1.2×10^7	N/m
μ	Friction coefficient	0.5	
ρ	Density	2.5	g/cm^3
Δt	Time interval	10^{-5}	s
V_d	Verlet distance	0.5	cm
G	Shear modulus	2×10^3	MPa
ν	Poisson's ratio	0.2	

5. Conclusions

An advanced numerical method, BSEM, has been developed for the simulation of the particle stress state, which will be used to model particle breakage in the future. It combines the advantages of the boundary element method and the spheropolygon element method. BSEM can effectively capture both the geometrical irregularity of particles and their interaction forces. The sub-particle stress can be accurately calculated using only meshes at the particle's boundary.

Performance analysis using the Brazilian test indicates that the stress fields provided by BSEM are more accurate and computationally less expensive than FEM-related methods for low-DoF calculation of linear elasticity. A domain mesh is not required, and thus breakage interfaces and the damage zone are not limited or affected by pre-defined elements. It has been pointed out that the computational time of the BSEM is considerably increased if a high-resolution stress field is calculated. However, a full stress field does not need to be calculated to simulate particle breakage. The stress state only needs to be considered at certain areas or lines on a particle, while the continuity and accuracy can still be maintained by using fundamental solutions in a better fashion than the shape function of FEM, which produces stress discontinuities. In the former case, the computational time for a high-resolution stress field can be largely reduced or even neglected.

Parametric studies using BSEM demonstrate advantages over existing formulas that are widely used in numerical methods by showing the discrepancy between the actual stress state and the results produced by the oversimplified assumptions that are generally applied in simulations. An advanced understanding of particle breakage requires accurate sub-particle stress fields. Further simulations of the uniaxial

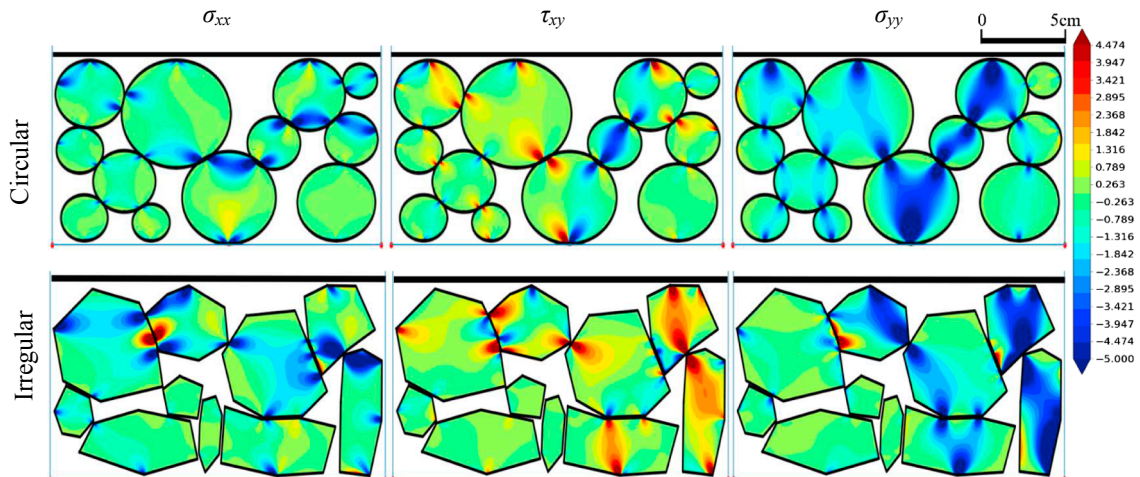


Fig. 16. Sub-particle stress of the circular and irregular particles (Units: MPa).

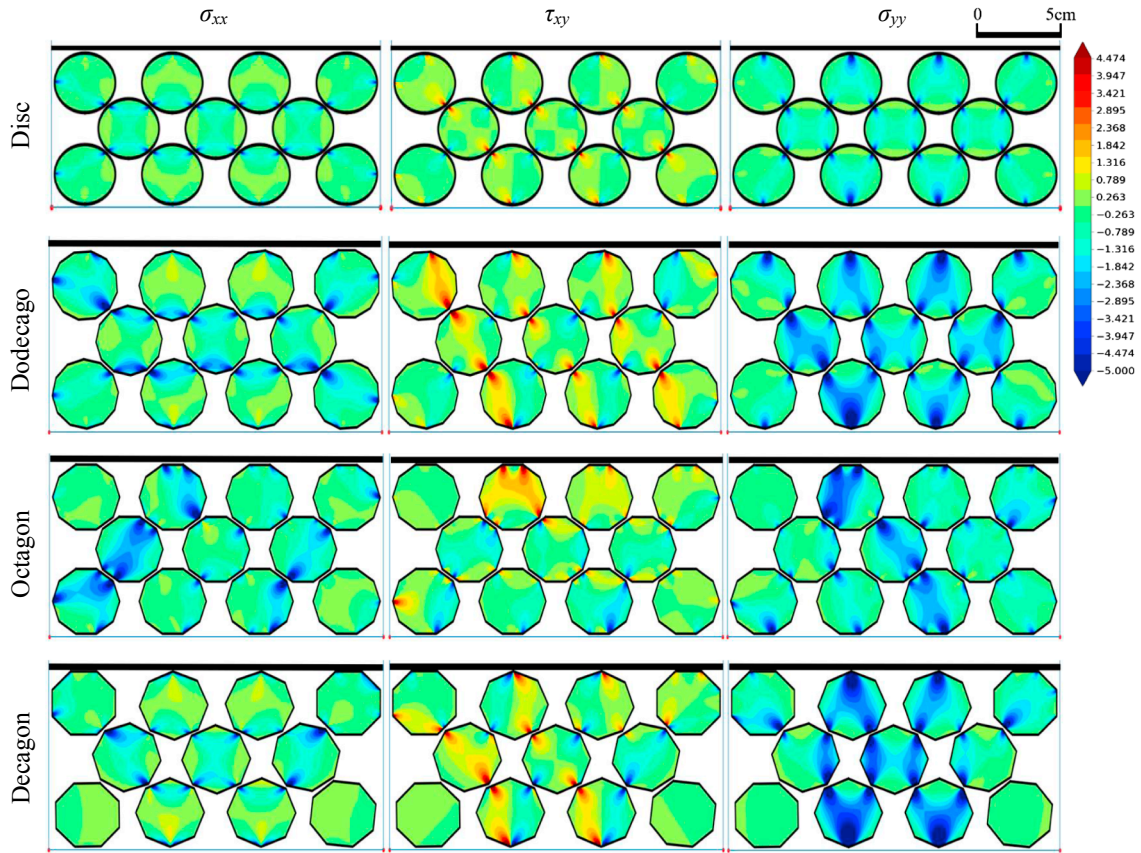


Fig. 17. Sub-particle stress components in discs and angular (dodecagon, decagon, and octagon) particles (Units: MPa).

loading test demonstrate that BSEM can provide insightful information on the sub-particle stress distribution. It indicates that decreases in the circularity of particles increase the heterogeneity of the force chain and hence cause high tensile stress areas in certain particles. This effect will lead to more breakage in the granular material and shows a lower breakage resistance for more angular particles.

It can be concluded that BSEM overcomes most of the existing problems in numerical methods for evaluation of the stress states of

particles. It provides a superior balance of accuracy and efficiency. BSEM has a great potential to instruct stress-based methods to simulate breakage interfaces and further improve the replacement strategy for broken particles.

While our method is still a one-way coupling method that does not include particle breakage, the analysis of the sub-particle stress provides important insights into the roles of particle aspect ratio and co-ordination number in particle breakage. The full-coupling method

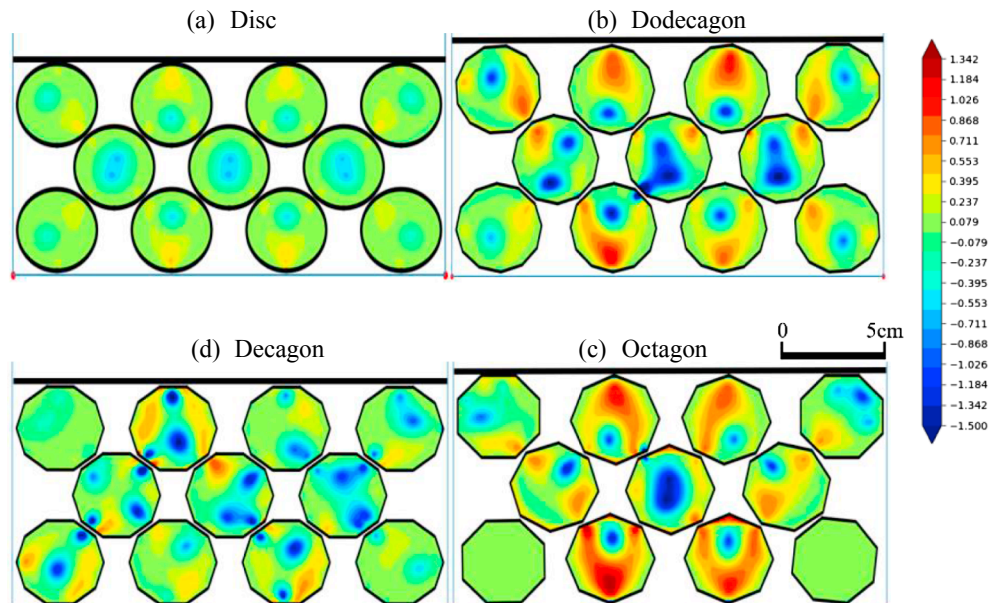


Fig. 18. Sub-particle maximum principal stress of the particle models with different circularity (Units: MPa).

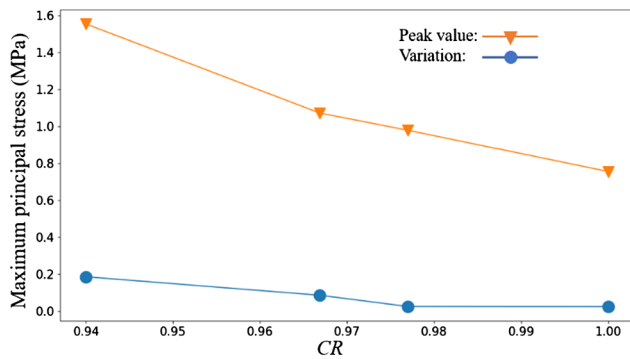


Fig. 19. The changes of peak value (PV) and averaged variation of the maximum principal stress (AVM) of the whole packing with different circularity.

requires the incorporation of failure criteria and energetic principles to replace the broken particle, and this will be discussed in future publications. We anticipate that BSEM will make a good contribution to the study of particle breakage in granular materials.

Acknowledgments

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